Duality Normalization for the Loeb-Rosser Functional Equation

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Abstract

Pass 66 refines the Pass 65 finite Verdier-dual equation

$$\mathbb{D}(\epsilon_S) = -\epsilon_S^\vee.$$

The finite matrix equation is stable, but the arithmetic-site lift depends on the choice of duality. This pass separates the finite-prime character-dual statement from the all-prime restricted-product problem.

Main Result

The unshifted $\mathbb{Z}$-linear dual is not the degree-preserving duality for the local cyclic layers:

$$\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n, \mathbb{Z}) = 0, \qquad \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n, \mathbb{Z}) \cong \mathbb{Z}/n.$$

Thus $R\mathrm{Hom}_{\mathbb{Z}}(-, \mathbb{Z})$ sees the local layers only after a cohomological shift.

The finite-prime normalization is character duality

$$D_{\mathrm{ch}}(A) = \mathrm{Hom}(A, \mathbb{Q}/\mathbb{Z}),$$

for which

$$D_{\mathrm{ch}}(\mathbb{Z}/n) \cong \mathbb{Z}/n.$$

For finite S, products and direct sums agree, so the Pass 65 signed transpose calculation gives

$$D_{\mathrm{ch}}(\epsilon_S) = -\epsilon_S^\vee.$$

For $S = \mathbb{P}$, the naive bare product fails: continuous characters of an infinite product have finite support, so the dual is a direct sum of local characters. The full adelic statement must therefore be formulated in a restricted-product / locally compact abelian setting.

Verification

The checker `code/scripts/check-pass66.py` generated `artifacts/reports/pass66-duality-normalization-scheme-lift-check.json`.

Verified facts:

- finite cyclic layers have trivial $\mathrm{Hom}(-, \mathbb{Z})$ but nontrivial $\mathrm{Ext}^1(-, \mathbb{Z})$ of the expected order;
- character duality preserves $\mathbb{Z}/p^k$ for $p = 2, 3, 5, 7$ and $k = 1, \dots, 4$;
- finite boundary matrices dualize by signed transpose and square back;
- finite product/direct-sum orders agree;
- finite-prefix support counts witness the infinite product/direct-sum obstruction.

Remaining Gap

Pass 67 should define the restricted-product coefficient for the full prime spectrum and check the global sign of the boundary class in that topological category.